

Working with an Amazon EC2 Instance

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Purpose

Learning more about the Amazon networking program, specifically the Amazon EC2 Instance. This included learning how to launch, configure, edit, and change an instance.

Lab Walkthrough (with Documentation)

Step 1: Configure Instance Details

Step 3: Configure Instance Details

▼ Advanced Details

Enclave ⓘ
☐ Enable

Metadata accessible ⓘ
Enabled

Metadata version ⓘ
V1 and V2 (token optional)

Metadata token response hop limit ⓘ
1

User data ⓘ
☒ As text ☐ As file ☐ Input is already base64 encoded

```
#!/bin/bash
yum -y install httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello Earthling, Take me to your leader!
</h1></html>' >/var/www/html/index.html
```

Cancel

Previous

Review and Launch

Next: Add Storage

Step 2: Add a Tag

Step 5: Add Tags

A tag consists of a case-sensitive key-value pair. For example, you could define a tag with key = Name and value = Webserver.

A copy of a tag can be applied to volumes, instances or both.

Tags will be applied to all instances and volumes. [Learn more](#) about tagging your Amazon EC2 resources.

Key (128 characters maximum)	Value (256 characters maximum)	Instances ⓘ	Volumes ⓘ	Network Interfaces ⓘ	
Name	BitBeat WebServer	☒	☒	☒	✕

Add another tag (Up to 50 tags maximum)

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[Step 3: Configure a Security Group](#)

Step 6: Configure Security Group

Assign a security group: ☒ Create a **new** security group

☐ Select an **existing** security group

Security group name:

Webserver SG

Description:

Security group for my web server

Type i	Protocol i	Port Range i	Source i
SSH v	TCP	22	Custom v 0.0.0.0/0
<button>Add Rule</button>			

[Step 4: Launch Instance](#)

Select an existing key pair or create a new key pair ×

A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

Proceed without a key pair v

☒ I acknowledge that I will not be able to connect to this instance unless I already know the password built into this AMI.

[Cancel](#)

[Launch Instances](#)

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Step 5: View/Verify your Instance

Instances (1) [Info](#)

[Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[Instance state: running](#) [Clear filters](#)

< 1 > [Settings](#)

☐	Name	Instance ID	Instance state	Instance type
☐	BitBeat Web...	i-00b79472a46ad0ae4	Running	t2.micro

Step 6: Use Public IP Address in a New Tab (Should Not Work at this Stage)

← → ↻ 🏠 ⚠ Not secure | 18.234.164.195 ☆ 🌐 ⋮

Apps Managed bookmarks C Clever | Portal 📺 Carnegie Learning... » | 📁 Other bookmarks

Bellevue School District 405

18.234.164.195 cannot be reached

Error

Reasons for this error message include:

- the website you're trying to reach doesn't exist or is broken.
- the web address was mistyped.
- you clicked on a link with an error.
- there is a problem with your Internet connection.

You are logged in as BSD\s-hwangg (IP Address: 24.17.215.52).

[If your error is "filtering error" please reboot. If this site is needed for school work click here to fill out a request to unblock include the blocked URL NOT the localhost page](#)

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Step 7: Create a Rule

Inbound rules [Info](#)

Inbound rule 1 Delete

Type [Info](#)

HTTP ▼

Port range [Info](#)

80

Source [Info](#)

0.0.0.0/0 ✕

::/0 ✕

Protocol [Info](#)

TCP

Source type [Info](#)

Anywhere ▼

Description - optional [Info](#)

Add rule

Step 8: Verify your Public IP

  54.89.224.144 

Hellow Earthling, Take me to your leader!

Step 9: Stop Instance

Instances (1) [Info](#)



Connect

Instance state ▼

Actions ▼

Launch instances ▼

 Filter instances

< 1 >



☐	Name ▼	Instance ID	Instance state ▼	Instance type ▼
☐	BitBeat WebSe...	i-0765df31947691800	 Stopped 	t2.micro

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Step 10: Change Instance Type

Change instance type [Info](#)
You can change the instance type only if the current instance type and the instance type that you want are compatible.

Instance ID
 i-0765df31947691800 (BitBeat WebServer)

Current instance type
t2.micro

Instance type

t2.small ▼

☐ EBS-optimized
EBS-optimized is not supported for this instance type

Cancel

Apply

Step 11: Verify your New Public IP



 54.227.83.23



Hellow Earthling, Take me to your leader!

Problems

One issue I had was when I tried accessing the website or public IP address that was supposed to work, it would not load. I double checked my code for the website and then my rules and could see no issues. I contacted a friend to compare labs. Everything in each looked the same from our screenshots, so we tried finding other solutions. A couple suggestions she made were to not use my school computer and use a different device, add a rule for HTTPS as well even though it is not included in the instructions, look at the different parts of the details under the instance, and more. We could not figure out what the issue was despite trying many things. In class however, one student who previously could not access the website as well said a couple solutions were to either use Firefox rather than Chrome, or to type http:// in front of your public IP address in Chrome. The latter worked and I was able to access my public IP address.

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